
Special Topics in Software Technology

Instructions for preparing the assignment

Winter semester of academic year 2023-2024

1. Subject of the work

You will need to design and develop a commercial (business) CRUD application, three (3) levels (front-end, business logic, database) on a small scale. By the term "small scale" we mean that it is not necessary to deal with all the possible functions of such an application but to choose 3-4 of those that are important, in your opinion. Besides, the aim of the work is not only the software that will be produced but also the development methodology.

2. Development methodology

You will be divided into teams of three (3) members maximum. Team registration is done here, [\[registration file \]](#), and you should fill in all the columns of the file. The design and development should follow the principles of agile development, i.e. at a minimum, the requirements will be in user form stories, there should be a product backlog and a board for each sprint. Sprints will have a duration of 15 days and it would be good if each user story could be completed in one sprint.

Tools you can use for project management (in their free versions) are:
indicatively the following:

• [Trello](#)

• [Jira](#)

• [Azure DevOps¹](#)

The start date of the job is Monday 20/11. The end date is the end of examination. In the online tools you will use, you must provide access to prezerak@uniwa.gr so that I can check the progress of your work.

¹ You have free access to all Microsoft tools, in the Azure market place, through your academic account.

3. Repo

As a version control system, you will use [Git](#). A satisfactory solution if you don't want to you work from the command line, it is the [SourceTree windowed environment](#). The code should uploaded to a web repo (preferably [GitHub](#), [Gitlab](#) or similar). Since the repo is private, you should give access to prezerak@uniwa.gr so I can check the progress of your work.

4. System architecture

As mentioned above, your application will have three layers (front-end, business logic, database). The constraints you should follow are (a) The front-end will communicate with the business logic using RESTful web services (b) the business logic should be based on an object-oriented language programming (c) the database must be relational and (d) the business logic must communicate with the database via ORM. Otherwise, you can choose the software stack of your choice.

5. Front-end

You have the following alternatives:

- (i) Based on what you have learned in our Department, the easiest thing for you is probably an application ~~Java with Swing desktop~~. For web service calls you have various options such as `URLConnection`, `HttpClient API` (Java 11 and later), [OkHttp](#) etc.
- (ii) Alternatively, you can develop a web front-end with the web pages being server-side rendered (e.g. Java servlets, PHP, MVC, Razor) and calls to web services are made, abusive², with javascript, perhaps with the help of a library like JQuery.
- (iii) Finally, you can develop a SPA front-end using a web framework such as Angular, Vue.js, React, Blazor etc.

For the needs of the labor market, the preferable option is (iii), but it requires a lot of ~~effort based on~~ the knowledge you have gained so far from our Department. Whatever choice you make will ~~not~~ will affect the score.

² "Abusive" because server-side rendering can work without web services.



6. Backend

It will be in the form of an API in 3 Layers (controllers, businesslogic, data) as we will say in the lesson. It should
you must use Dependency Injection.

7. Database

You can choose whichever relational database you prefer. Examples include: _____

• Ms Sql Server Express / Enterprise

• Oracle

• PostgreSQL

• MariaDB

• MySQL

8. ORM

It would be good to communicate with the database via ORM3 . There are various ORMs available.

depending on the software stack and database, such as (indicatively): _____

• .Net1 Entity Framework

• Java Hibernate

• Ruby ActiveRecord

• Python Django ORM

• PHP Doctrine

9. Authorization / Authentication

Your application should have a basic AA based on username / password. Since _____

If possible, try to implement role-based access to functions.

access). I recommend the JWT approach but it is not mandatory.

³ If you can't do that then you can use SQL but you should have exhausted any other possibility. In such a case, there will be a small (negative) impact on the score.

10. Testing

In general, CRUD applications do not need unit-testing. Also, at the time of writing this the text, you have not delved particularly into how unit tests are designed and executed. Therefore you will develop some basic automated integration tests at the end of the semester (abuse _____ because normally we have to test before each sprint is completed).

Usually the frameworks used for unit testing are also used for integration.

testing. Examples include:

• .Net1	xUnit / NUnit / MsTest
• Java	Junit
• Ruby	Built-in
• Python	PyTest
• PHP	PHPUnit

11. Evaluation

Field	Units
Existence and use of a code repository	1
Existence of a tool for tracking sprints	2
Conducting development based on sprints	2
Using Dependency Injection	1
Using ORM	1
Existence of integration tests	2
Code Format / UI Look & feel	1